1) a)
Reflection in the x-axis

1) b)
Reflection in the line $y = x$

1) c)
Reflection in the y-axis

1) d)
Reflection in the line $y = -x$

2) a)
$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

2) b)
$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

2) c)
$\mathbf{A}^3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

2) d)
$\mathbf{A}^{2025} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

3)
The matrix represents a Reflection in the y-axis. So the image will have the same area as the original shape. So the area of the shape's image is 4 units^2

4) a)
$(3, 5) \mapsto (3, -5)$ is a reflection in the x-axis
$\therefore \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

4) b)
$(-6, -2) \mapsto (6, -2)$ is a reflection in the y-axis
$\therefore \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

4) c)

$(1,9) \mapsto (-9,-1)$ is a reflection in the line $y = -x$

$\therefore \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

4) d)

$(-4,-7) \mapsto (-7,-4)$ is a reflection in the line $y = x$

$\therefore \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

5)

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 8 & 9 & 10 \\ 1 & 3 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 3 & -3 \\ 8 & 9 & 10 \end{pmatrix}$

So the coordinates of the vertices of T' are $(1,8)$, $(3,9)$ and $(-3,10)$

6)

$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} -1 & 6 & 11 & -1 \\ 5 & 5 & -3 & -3 \end{pmatrix} = \begin{pmatrix} -1 & 6 & 11 & -1 \\ -5 & -5 & 3 & 3 \end{pmatrix}$

So the coordinates of the vertices of X' are $(-1,-5)$, $(6,-5)$, $(11,3)$ and $(-1,3)$

7) a)

First reflect the point in the line $y = x$:

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ -5 \end{pmatrix} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}$

The vector from the origin to the closest point on the line is $\begin{pmatrix} -2.5 \\ 2.5 \end{pmatrix}$

So the position vector of $(3,-5)$ when reflected in the line $y = x + 5$ is:

$\begin{pmatrix} -5 \\ 3 \end{pmatrix} + 2\begin{pmatrix} -2.5 \\ 2.5 \end{pmatrix} = \begin{pmatrix} -10 \\ 8 \end{pmatrix}$

$\therefore (-10,8)$

7) b)

First reflect the point in the line $y = -x$:

$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ -5 \end{pmatrix} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$

The vector from the origin to the closest point on the line is $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$

So the position vector of $(3,-5)$ when reflected in the line $y = 8 - x$ is:

$\begin{pmatrix} 5 \\ -3 \end{pmatrix} + 2\begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} 13 \\ 5 \end{pmatrix}$

$\therefore (13,5)$

8) a)

$$\begin{pmatrix}\cos(2\times30°) & \sin(2\times30°)\\ \sin(2\times30°) & -\cos(2\times30°)\end{pmatrix}=\begin{pmatrix}\cos 60° & \sin 60°\\ \sin 60° & -\cos 60°\end{pmatrix}=\begin{pmatrix}\frac{1}{2} & \frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{pmatrix}$$

8) b)

$$\begin{pmatrix}\cos(2\times60°) & \sin(2\times60°)\\ \sin(2\times60°) & -\cos(2\times60°)\end{pmatrix}=\begin{pmatrix}\cos 120° & \sin 120°\\ \sin 120° & -\cos 120°\end{pmatrix}=\begin{pmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix}$$

8) c)

$$\begin{pmatrix}\cos(2\times15°) & \sin(2\times15°)\\ \sin(2\times15°) & -\cos(2\times15°)\end{pmatrix}=\begin{pmatrix}\cos 30° & \sin 30°\\ \sin 30° & -\cos 30°\end{pmatrix}=\begin{pmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ \frac{1}{2} & -\frac{\sqrt{3}}{2}\end{pmatrix}$$

8) d)

$$\begin{pmatrix}\cos(2\times75°) & \sin(2\times75°)\\ \sin(2\times75°) & -\cos(2\times75°)\end{pmatrix}=\begin{pmatrix}\cos 150° & \sin 150°\\ \sin 150° & -\cos 150°\end{pmatrix}=\begin{pmatrix}-\frac{\sqrt{3}}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}$$

9) a)

$\tan\theta=\sqrt{2}\Rightarrow\theta=\arctan\sqrt{2}=54.7°$ (3 SF)

$$\begin{pmatrix}\cos\left(2\arctan\sqrt{2}\right) & \sin\left(2\arctan\sqrt{2}\right)\\ \sin\left(2\arctan\sqrt{2}\right) & -\cos\left(2\arctan\sqrt{2}\right)\end{pmatrix}=\begin{pmatrix}-\frac{1}{3} & \frac{2\sqrt{2}}{3}\\ \frac{2\sqrt{2}}{3} & \frac{1}{3}\end{pmatrix}=\begin{pmatrix}-0.333 & 0.943\\ 0.943 & 0.333\end{pmatrix}\text{ (3SF)}$$

9) b)

$\tan\theta=3\Rightarrow\theta=\arctan 3=71.6°$ (3 SF)

$$\begin{pmatrix}\cos(2\arctan 3) & \sin(2\arctan 3)\\ \sin(2\arctan 3) & -\cos(2\arctan 3)\end{pmatrix}=\begin{pmatrix}-\frac{4}{5} & \frac{3}{5}\\ \frac{3}{5} & \frac{4}{5}\end{pmatrix}=\begin{pmatrix}-0.8 & 0.6\\ 0.6 & 0.8\end{pmatrix}$$

9) c)

$\tan\theta=\frac{1}{8}\Rightarrow\theta=\arctan\frac{1}{8}=7.13°$ (3 SF)

$$\begin{pmatrix}\cos\left(2\arctan\frac{1}{8}\right) & \sin\left(2\arctan\frac{1}{8}\right)\\ \sin\left(2\arctan\frac{1}{8}\right) & -\cos\left(2\arctan\frac{1}{8}\right)\end{pmatrix}=\begin{pmatrix}\frac{63}{65} & \frac{16}{65}\\ \frac{16}{65} & -\frac{63}{65}\end{pmatrix}=\begin{pmatrix}0.969 & 0.246\\ 0.246 & -0.969\end{pmatrix}\text{ (3SF)}$$

9) d)

$\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \arctan\frac{1}{\sqrt{3}} = 30°$

$$\begin{pmatrix} \cos 60° & \sin 60° \\ \sin 60° & -\cos 60° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0.500 & 0.866 \\ 0.866 & -0.500 \end{pmatrix} \text{ (3SF)}$$

9) e)

$\tan\theta = -2 \Rightarrow \theta = 180° + \arctan(-2) = 117°$ (3 SF)

$$\begin{pmatrix} \cos(2(180° + \arctan(-2))) & \sin(2(180° + \arctan(-2))) \\ \sin(2(180° + \arctan(-2))) & -\cos(2(180° + \arctan(-2))) \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{pmatrix}$$

$$= \begin{pmatrix} -0.6 & -0.8 \\ -0.8 & 0.6 \end{pmatrix}$$

9) f)

$\tan\theta = -\frac{1}{\sqrt{2}} \Rightarrow \theta = 180° + \arctan\left(-\frac{1}{\sqrt{2}}\right) = 145°$ (3 SF)

$$\begin{pmatrix} \cos\left(2\left(180° + \arctan\left(-\frac{1}{\sqrt{2}}\right)\right)\right) & \sin\left(2\left(180° + \arctan\left(-\frac{1}{\sqrt{2}}\right)\right)\right) \\ \sin\left(2\left(180° + \arctan\left(-\frac{1}{\sqrt{2}}\right)\right)\right) & -\cos\left(2\left(180° + \arctan\left(-\frac{1}{\sqrt{2}}\right)\right)\right) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{3} & -\frac{2\sqrt{2}}{3} \\ -\frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} 0.333 & -0.943 \\ -0.943 & -0.333 \end{pmatrix} \text{ (3SF)}$$

9) g)

$\tan\theta = -\frac{1}{5} \Rightarrow \theta = 180° + \arctan\left(-\frac{1}{5}\right) = 169°$ (3 SF)

$$\begin{pmatrix} \cos\left(2\left(180° + \arctan\left(-\frac{1}{5}\right)\right)\right) & \sin\left(2\left(180° + \arctan\left(-\frac{1}{5}\right)\right)\right) \\ \sin\left(2\left(180° + \arctan\left(-\frac{1}{5}\right)\right)\right) & -\cos\left(2\left(180° + \arctan\left(-\frac{1}{5}\right)\right)\right) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{12}{13} & -\frac{5}{13} \\ -\frac{5}{13} & -\frac{12}{13} \end{pmatrix} = \begin{pmatrix} 0.923 & -0.385 \\ -0.385 & -0.923 \end{pmatrix} \text{ (3SF)}$$

9) h)

$\tan\theta = -\frac{1}{5\sqrt{3}} \Rightarrow \theta = 180° + \arctan\left(-\frac{1}{5\sqrt{3}}\right) = 173°$ (3 SF)

$$\begin{pmatrix} \text{Cos}\left(2\left(180° + \arctan\left(-\frac{1}{5\sqrt{3}}\right)\right)\right) & \sin\left(2\left(180° + \arctan\left(-\frac{1}{5\sqrt{3}}\right)\right)\right) \\ \sin\left(2\left(180° + \arctan\left(-\frac{1}{5\sqrt{3}}\right)\right)\right) & -\cos\left(2\left(180° + \arctan\left(-\frac{1}{5\sqrt{3}}\right)\right)\right) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{37}{38} & -\frac{5\sqrt{3}}{38} \\ -\frac{5\sqrt{3}}{38} & -\frac{37}{38} \end{pmatrix} = \begin{pmatrix} 0.974 & -0.228 \\ -0.228 & -0.974 \end{pmatrix}$$ (3SF)

10) a)

$\cos 2\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{75°, 105°\}$

$\sin 2\theta = \frac{1}{2} \Rightarrow \theta = \{15°, 75°\}$

So $\theta = 75°$

So the mirror line has equation $y = (\tan 75°)x \Rightarrow y = \left(2 + \sqrt{3}\right)x$

$\Rightarrow y = 3.73x$ (3 SF)

10) b)

$\cos 2\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{75°, 105°\}$

$\sin 2\theta = -\frac{1}{2} \Rightarrow \theta = \{105°, 165°\}$

So $\theta = 105°$

So the mirror line has equation $y = (\tan 105°)x \Rightarrow y = \left(-2 - \sqrt{3}\right)x$

$\Rightarrow y = -3.73x$ (3 SF)

10) c)

$\cos 2\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{7.5°, 172.5°\}$

$\sin 2\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{7.5°, 82.5°\}$

So $\theta = 7.5°$

So the mirror line has equation $y = (\tan 7.5°)x \Rightarrow y = 0.132x$ (3 SF)

10) d)

$\cos 2\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{37.5°, 142.5°\}$

$\sin 2\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{37.5°, 52.5°\}$

So $\theta = 37.5°$

So the mirror line has equation $y = (\tan 37.5°)x \Rightarrow y = 0.767x$ (3 SF)

11)

Midpoint of $(4, 6)$ and $\left(3 + 2\sqrt{3}, 2 - 3\sqrt{3}\right)$ is $\left(\frac{7}{2} + \sqrt{3}, 4 - \frac{3}{2}\sqrt{3}\right)$

Gradient from the origin to the midpoint is $\frac{4-\frac{3}{2}\sqrt{3}}{\frac{7}{2}+\sqrt{3}} = 2 - \sqrt{3}$

The angle the mirror line makes with the horizontal is $\theta = \arctan\left(2 - \sqrt{3}\right) = 15°$

$$\therefore \begin{pmatrix} \cos(2 \times 15°) & \sin(2 \times 15°) \\ \sin(2 \times 15°) & -\cos(2 \times 15°) \end{pmatrix} = \begin{pmatrix} \cos 30° & \sin 30° \\ \sin 30° & -\cos 30° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

12)

Midpoint of $(-18, 20)$ and $\left(9 + 10\sqrt{3}, 10 - 9\sqrt{3}\right)$ is $\left(-\frac{9}{2} + 5\sqrt{3}, 15 - \frac{9}{2}\sqrt{3}\right)$

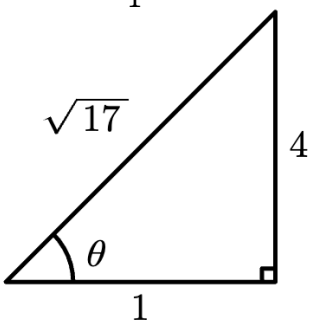
Gradient from the origin to the midpoint is $\frac{15-\frac{9}{2}\sqrt{3}}{-\frac{9}{2}+5\sqrt{3}} = \sqrt{3}$

The angle the mirror line makes with the horizontal is $\theta = \arctan\sqrt{3} = 60°$

$$\therefore \begin{pmatrix} \cos(2 \times 60°) & \sin(2 \times 60°) \\ \sin(2 \times 60°) & -\cos(2 \times 60°) \end{pmatrix} = \begin{pmatrix} \cos 120° & \sin 120° \\ \sin 120° & -\cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

13)

$\tan\theta = \frac{4}{1}$



$\Rightarrow \sin\theta = \frac{4}{\sqrt{17}}$ and $\cos\theta = \frac{1}{\sqrt{17}}$

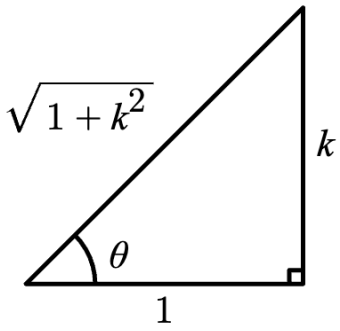
$\cos 2\theta = 2\cos^2\theta - 1 = 2\left(\frac{1}{\sqrt{17}}\right)^2 - 1 = -\frac{15}{17}$

$\sin 2\theta = 2\sin\theta\cos\theta = 2 \times \frac{4}{\sqrt{17}} \times \frac{1}{\sqrt{17}} = \frac{8}{17}$

$$\therefore \mathbf{A} = \begin{pmatrix} -\frac{15}{17} & \frac{8}{17} \\ \frac{8}{17} & \frac{15}{17} \end{pmatrix}$$

14)

$\tan\theta = k$



$\Rightarrow \sin\theta = \frac{k}{\sqrt{1+k^2}}$ and $\cos\theta = \frac{1}{\sqrt{1+k^2}}$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = \left(\frac{1}{\sqrt{1+k^2}}\right)^2 - \left(\frac{k}{\sqrt{1+k^2}}\right)^2 = \frac{1}{1+k^2} - \frac{k^2}{1+k^2} = \frac{1-k^2}{1+k^2}$

$\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{k}{\sqrt{1+k^2}}\right)\left(\frac{1}{\sqrt{1+k^2}}\right) = \frac{2k}{1+k^2}$

$\therefore \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & -\frac{1-k^2}{1+k^2} \end{pmatrix} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}$